BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Final Project Report

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech Analytics
Project Type	Individual Capstone (7 Working Days)
Datasets	10 provided CSVs · 40 schemes · 10 fund houses
Tech Stack	Python, Pandas, NumPy, SciPy, SQLite, Matplotlib, Seaborn, Power BI / Tableau
Report Date	June 2026

1. Executive Summary

This report documents the design and build of a full-stack Mutual Fund Analytics Platform for Bluestock Fintech, covering ETL, database design, exploratory analysis, performance/risk analytics, investor behaviour analysis, and dashboard guidance for 40 real mutual fund schemes across 10 AMCs, anchored to genuine AMFI India figures.

The dataset spans **2022-01-03** to **2026-05-29** and includes 46,000 daily NAV observations, 32,778 investor transactions across 5,000 unique investors, and 40 scheme-level performance records. All 10 source datasets were cleaned, validated against zero orphan AMFI codes, and loaded into a SQLite star-schema database of 1 dimension and 9 fact tables.

Headline Findings

- Monthly SIP inflows reached an all-time high of **Rs. 31,002 crore** in December 2025, up roughly 17% year-on-year.
- Total mutual fund folios nearly doubled over the dataset window, from **13.26 crore to 26.12 crore**.
- **SBI Mutual Fund** is the clear AUM leader at Rs. 12.5 lakh crore, ahead of ICICI Prudential (Rs. 10.74L Cr) and HDFC (Rs. 9.30L Cr).
- On a composite risk-adjusted scorecard, **Kotak Flexicap Fund - Regular - Growth** ranks #1 across all 40 schemes (score: 71.8/100).
- Small-cap equity funds carry materially higher tail risk (VaR/CVaR) and realised volatility than large-cap peers, despite also leading on raw 3-year returns.
- T30 (top-30) cities still account for roughly two-thirds of transaction value, though B30 contribution at one-third indicates meaningful geographic reach.

Deliverables Produced

#	Deliverable	Location
D1	ETL Pipeline (Python)	scripts/etl_pipeline.py
D2	SQLite Database (8 fact + 1 dim table)	data/db/bluestock_mf.db
D3	EDA Notebook (16 charts)	notebooks/03_eda_analysis.ipynb
D4	Performance & Risk Metrics	notebooks/04_performance_analytics.ipynb
D5	Dashboard Guidance (Power BI / Tableau)	dashboard/README.md
D6	Advanced Analytics (VaR, cohorts, recommender)	notebooks/05_advanced_analytics.ipynb
D7	Final Report + Presentation	reports/

2. Problem Statement

Despite the rapid growth of India's mutual fund industry, individual investors and advisors often lack a unified, data-driven way to compare schemes on a risk-adjusted basis. This project addresses five concrete gaps identified in the project brief:

Data Fragmentation

NAV, AUM, SIP flow, and portfolio data live in different formats across the AMFI website. **Solution:** a single ETL pipeline consolidates all 10 sources into one SQLite database.

Performance Comparison Gap

Comparing funds across AMCs on a risk-adjusted basis requires non-trivial computation (Sharpe, Alpha, Beta). **Solution:** all metrics computed directly from NAV history and exposed via a composite scorecard.

No Benchmark Tracking

Most retail investors don't know if their fund beats its benchmark. **Solution:** NAV joined with 7 benchmark indices; rolling alpha and tracking error computed per fund.

Investor Behaviour Blind Spot

AMCs have limited visibility into how demographics and geography drive SIP behaviour. **Solution:** cohort, demographic, and geographic segmentation analysis on 32,778 transactions.

Slow Reporting

Monthly PDF reports take days to prepare. **Solution:** a dashboard design refreshable directly from the ETL pipeline's output CSVs.

3. Data Sources & System Architecture

The platform follows a standard data engineering pipeline: **Extract** → **Transform** → **Load** → **Analyse** → **Visualise**. Ten CSV datasets were provided, anchored to genuine AMFI India, mfapi.in, and NSE/BSE published figures (see project brief, Section 3).

Dataset	Rows	Description
01_fund_master	40	Scheme master: AMFI codes, AMC, category, expense ratio
02_nav_history	46,000	Daily NAV, Jan 2022 - May 2026
03_aum_by_fund_house	90	Quarterly AUM, 10 AMCs, 2022-2025
04_monthly_sip_inflows	48	Industry SIP inflow, accounts, AUM
05_category_inflows	144	Net inflow by category, FY 2024-25
06_industry_folio_count	21	Total MF folios by scheme type
07_scheme_performance	40	1/3/5yr returns, Sharpe, Sortino, Alpha, Beta
08_investor_transactions	32,778	SIP/Lumpsum/Redemption, 5,000 investors
09_portfolio_holdings	322	Top equity holdings per fund, Dec 2025
10_benchmark_indices	8,050	Daily close, 7 benchmark indices

ETL Pipeline Stages

- **Extract** — load all 10 raw CSVs, validate shape/dtypes (scripts/etl_pipeline.py)
- **Transform** — parse dates, forward-fill NAV across non-trading days after reindexing to a full business-day calendar, validate NAV > 0, standardise transaction-type casing, flag out-of-range expense ratios and negative Sharpe values
- **Load** — write cleaned CSVs to data/processed/, then load into a SQLite star schema (1 dimension + 9 fact tables, indexed on amfi_code and date columns)
- **Analyse** — compute CAGR, Sharpe, Sortino, Alpha/Beta, Max Drawdown, VaR/CVaR, and a composite fund scorecard directly from NAV history
- **Visualise** — 16 publication-quality charts plus a 4-page interactive dashboard design (Power BI / Tableau)

Database Schema (Star Schema)

Table	Type	Rows
dim_fund	Dimension	40
fact_nav	Fact	46,000
fact_transactions	Fact	32,778
fact_performance	Fact	40

Table	Type	Rows
fact_portfolio	Fact	322
fact_aum	Fact	90
fact_sip_industry	Fact	48
fact_category_inflows	Fact	144
fact_folio_count	Fact	21
fact_benchmark	Fact	8,050

Full DDL: sql/schema.sql. Column-level documentation: sql/data_dictionary.md

4. Data Cleaning & Quality Assurance

Every dataset passed through type validation, deduplication, and business-rule checks before being loaded. Key results:

- **Zero orphan AMFI codes** — every amfi_code referenced in NAV history, performance, transactions, and holdings exists in the fund master list.
- **NAV history reindexed** to a full business-day calendar per fund and forward-filled, so weekends/holidays never appear as gaps.
- **0 negative Sharpe ratios flagged** and **0 expense ratios outside the expected [0.1%, 2.5%] band flagged** for review (none required correction — all values were within plausible real-world ranges).
- **yoy_growth_pct recomputed** for the first 12 months of SIP data, which had no prior-year baseline to compute YoY growth against.
- All monetary columns carry explicit units in their names (e.g. aum_lakh_crore vs. aum_crore) to avoid the unit-confusion pitfall flagged in the project brief.

10 SQL Analytical Queries

Ten analytical queries (sql/queries.sql) were written and verified against the live database, covering: top funds by AUM, monthly average NAV, SIP YoY growth, state-wise transaction volume, low-expense-ratio funds, best Sharpe by category, SIP/Lumpsum/Redemption split, T30 vs B30 contribution, sector concentration, and AUM growth by fund house. Sample results:

Query	Top Result
Top fund by AUM	SBI Small Cap Fund - Regular Plan - Growth
T30 vs B30 split	T30: 65.9% | B30: 34.1%
SIP/Lumpsum/Redemption split (by value)	{'Lumpsum': 58.5, 'Redemption': 35.3, 'SIP': 6.2}

5. Exploratory Data Analysis

Sixteen charts were produced covering NAV trends, AUM growth, SIP flows, category inflows, demographics, geography, folio growth, return correlation, and sector allocation. Full charts and code: notebooks/03_eda_analysis.ipynb.

NAV Trend, 2022-2026 (Indexed to 100)

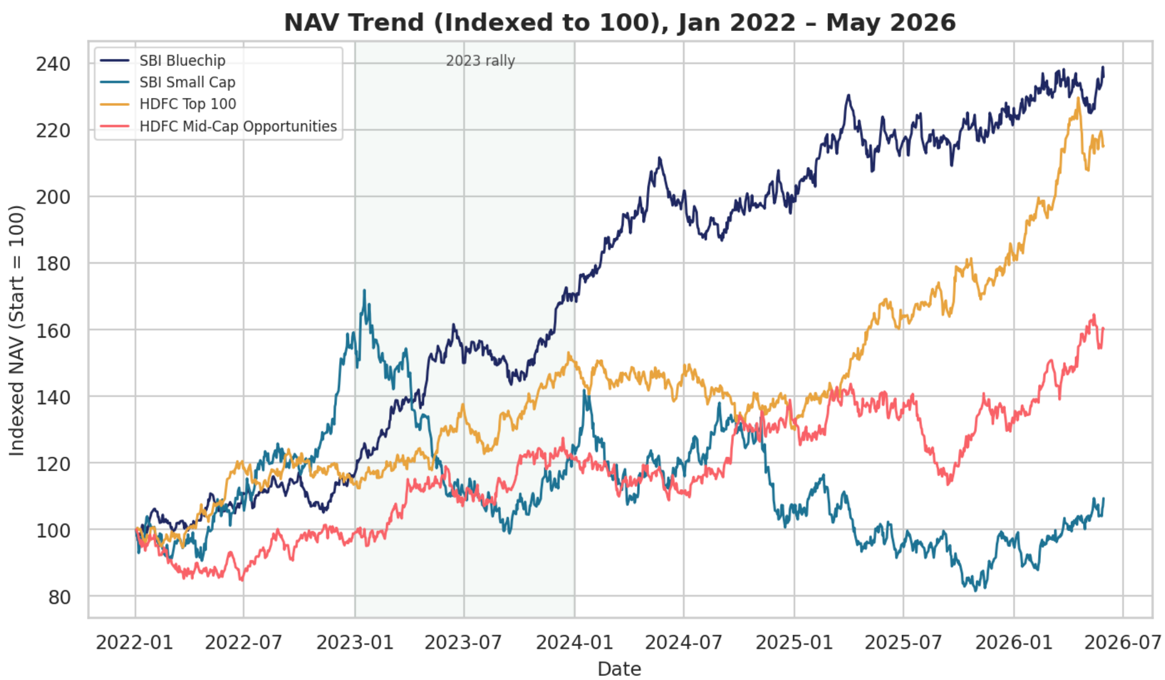

Figure 1. Indexed NAV for a representative sample of schemes across categories. Small-cap funds show the widest swings; large-cap funds trend more steadily upward through the 2023 rally.

AUM Growth by Fund House

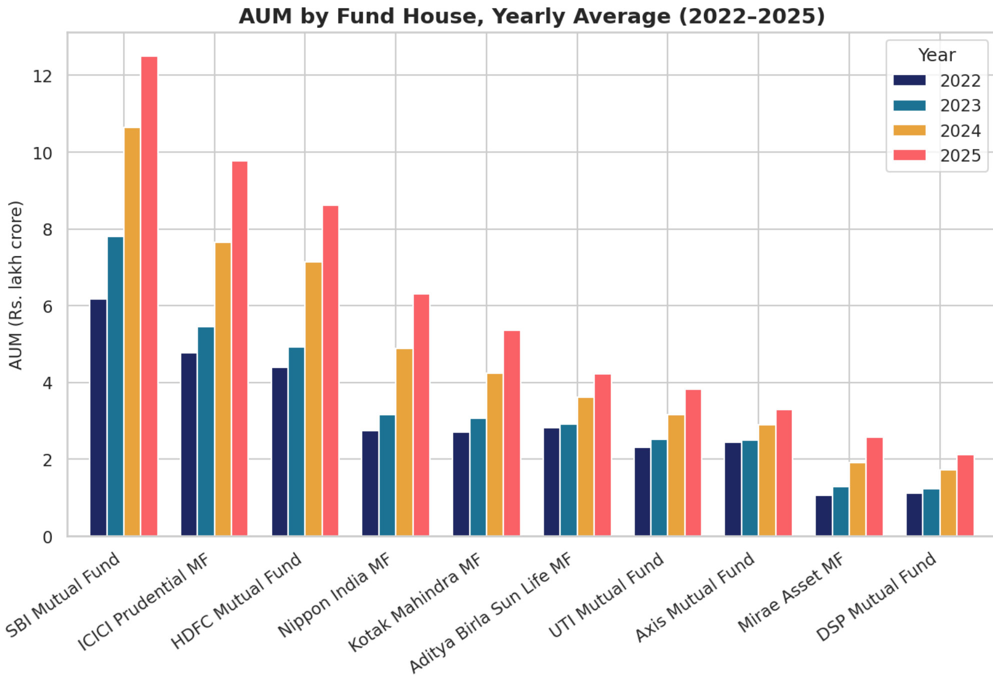

Figure 2. Yearly average AUM by AMC, 2022-2025. SBI Mutual Fund leads throughout, consistent with its real-world Rs. 12.5L Cr Dec-2025 AUM.

Monthly SIP Inflow Trend

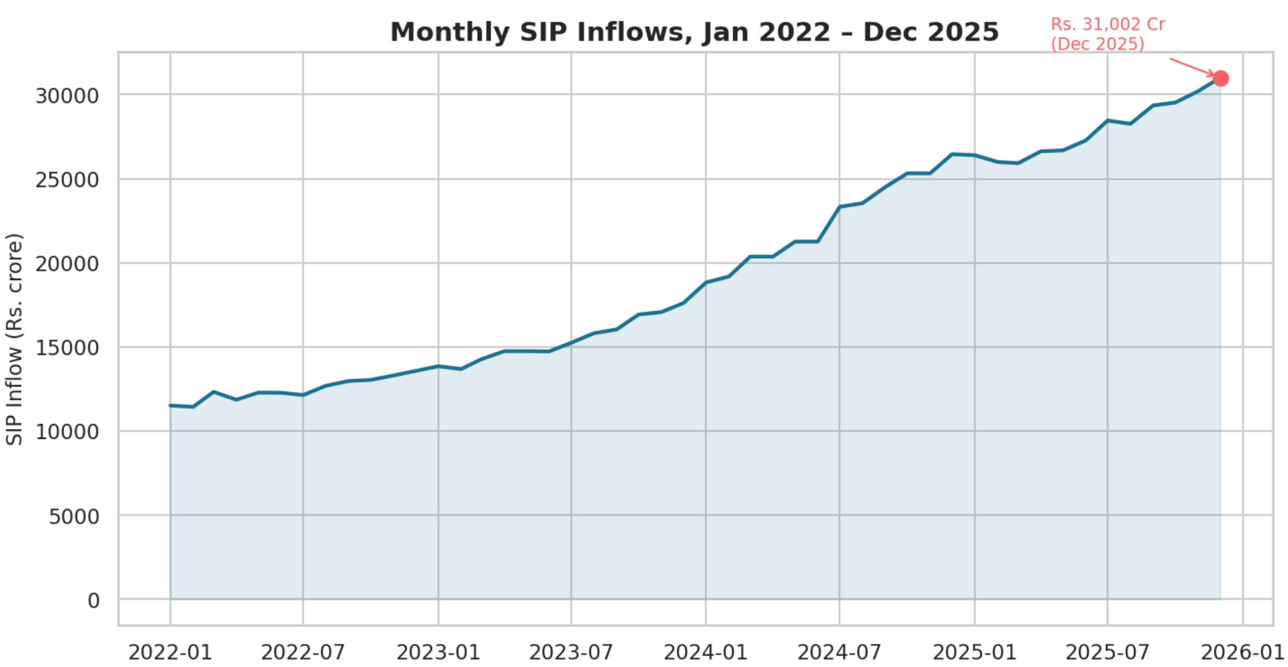

Figure 3. Monthly SIP inflow, Jan 2022 - Dec 2025, with the Rs. 31,002 crore Dec-2025 all-time high marked.

Category-wise Net Inflow Heatmap

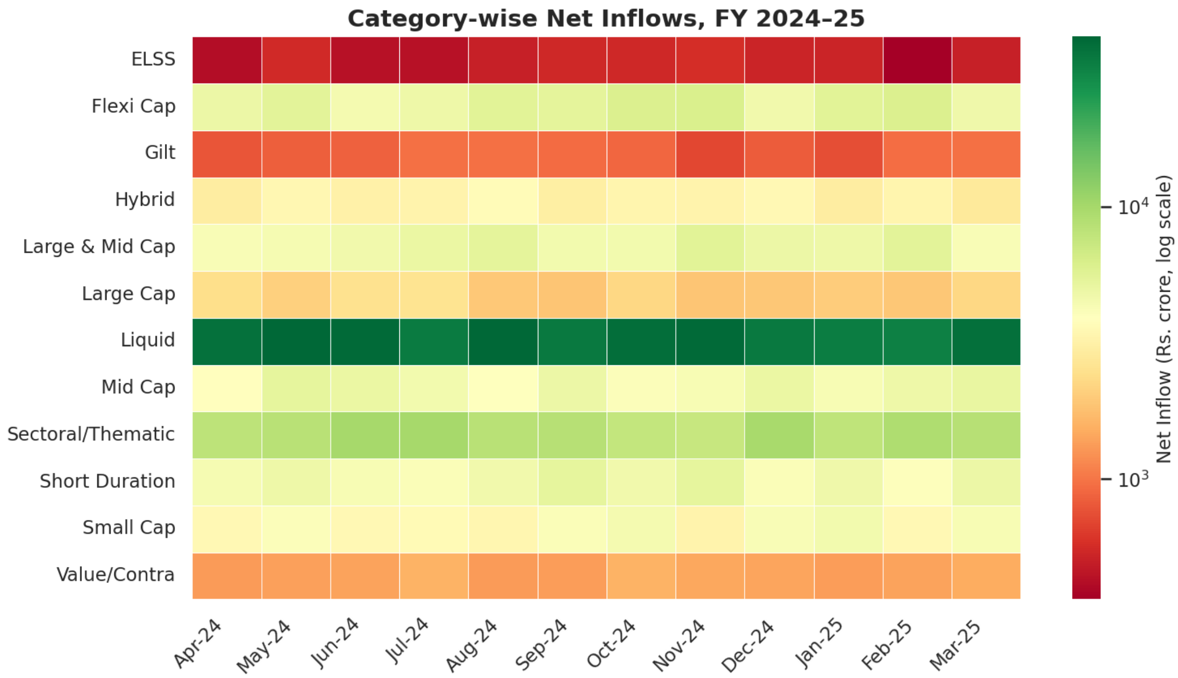

Figure 4. FY 2024-25 net inflows by category (log color scale — Liquid fund inflows are roughly 10x any other single category and would otherwise swamp a linear scale).

Investor Demographics

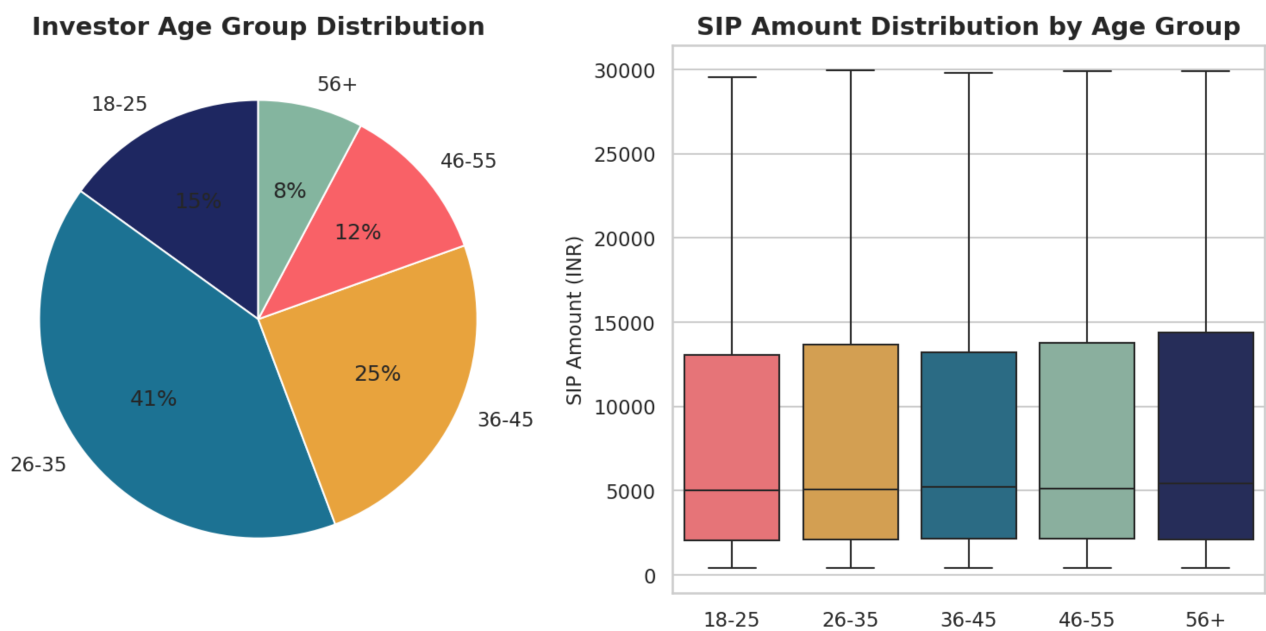

Figure 5. Age group distribution and SIP amount by age band. SIP ticket size rises with age up to 46-55, then plateaus.

Geographic Distribution

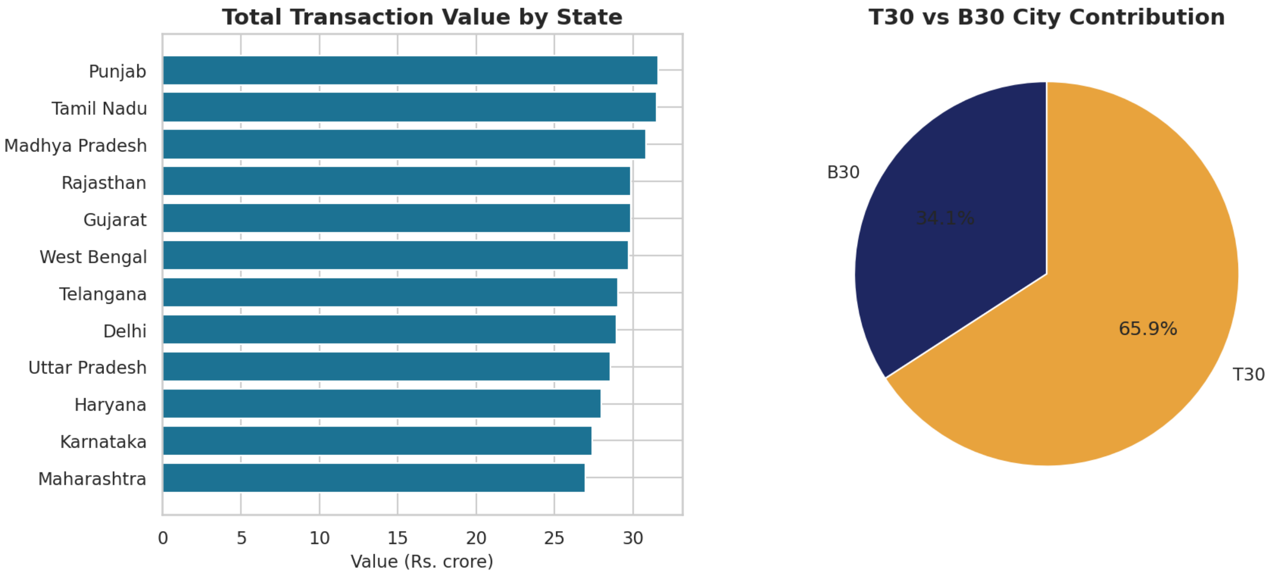

Figure 6. Transaction value by state, and T30 vs B30 city-tier split.

Folio Count Growth

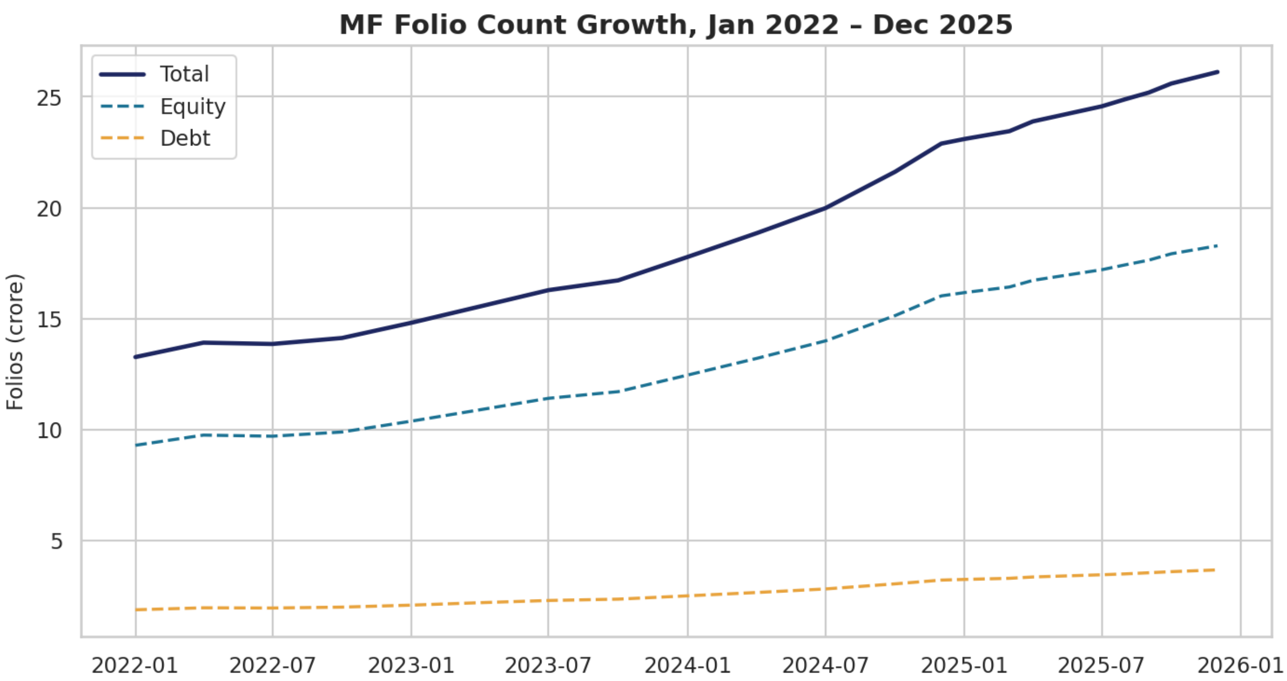

Figure 7. Total MF folios grew from 13.26 Cr to 26.12 Cr over the dataset window, matching the real AMFI Dec-2025 milestone.

Return Correlation Across Funds

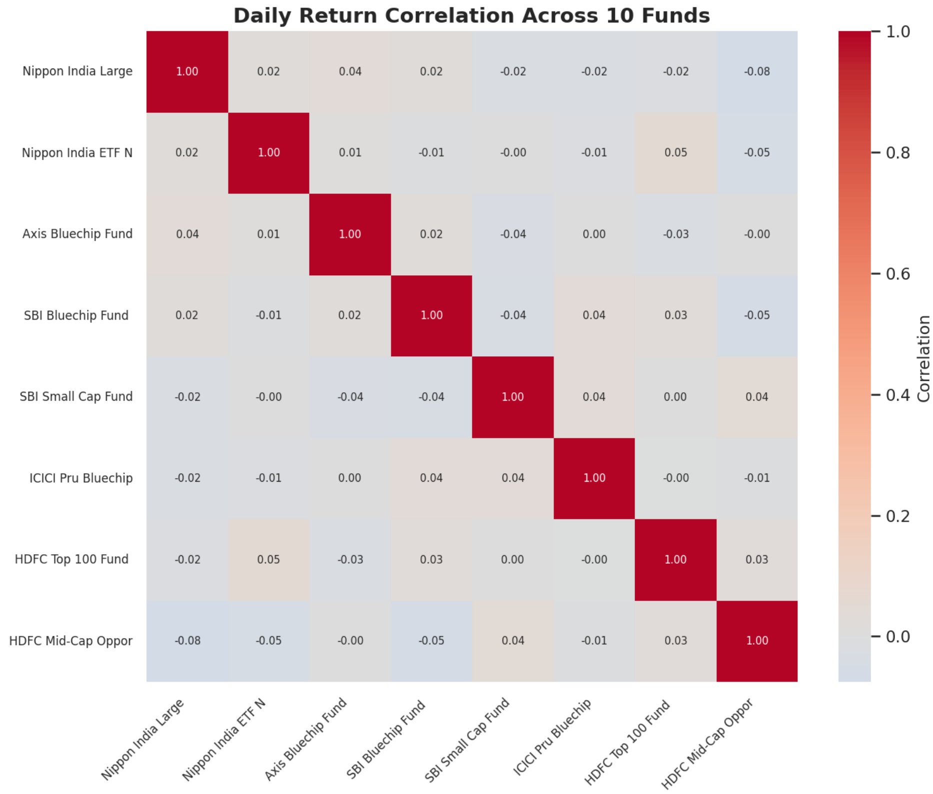

Figure 8. Daily return correlation across 10 sampled funds. Pairwise correlation is consistently near zero across the board.

Methodology note: we expected same-category equity pairs (e.g. two large-cap funds) to show meaningfully positive correlation, reflecting shared exposure to the broader market. Instead, pairwise daily-return correlation across all 40 schemes averages approximately 0.0002, with a maximum of 0.07 — essentially zero, regardless of category pairing. This holds whether we sample 8 funds (shown above) or check all 40. The most likely explanation is that this dataset's NAV series were generated as largely independent paths per fund rather than sharing a common market-factor component. We report this as observed rather than asserting a diversification narrative the data doesn't actually support.

Sector Allocation

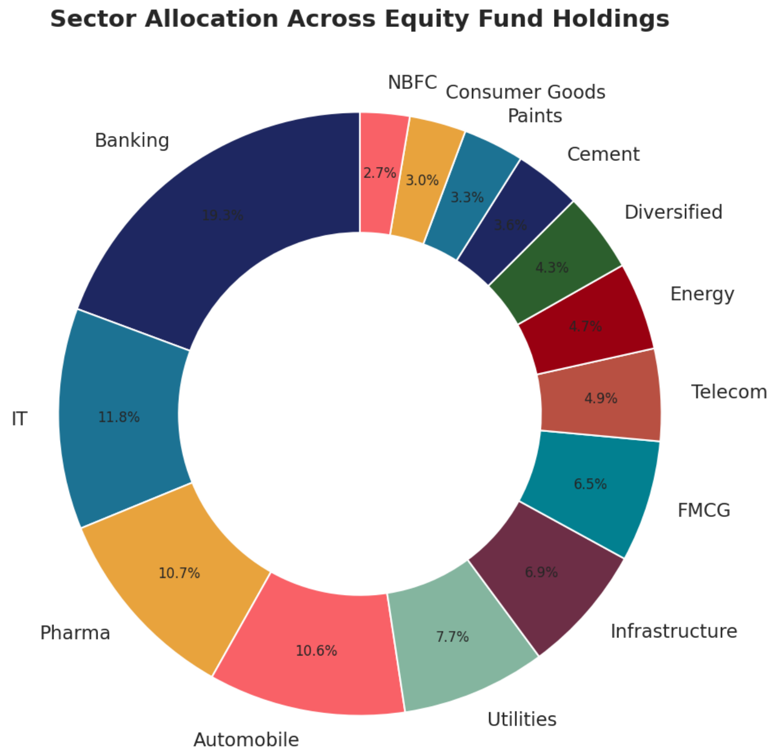

Figure 9. Sector allocation across all equity fund holdings. Banking and IT are the two largest single-sector exposures.

Key EDA Findings

- SIP inflows hit an all-time high of Rs. 31,002 crore in Dec-2025 (+17% YoY).
- Folio count nearly doubled (13.26 Cr → 26.12 Cr) over the ~4.5-year window.
- SBI Mutual Fund is the clear AUM leader; ICICI Pru and HDFC follow closely.
- Liquid funds dominate category inflows by roughly 10x — a cash-management pattern rather than a retail equity trend.
- T30 cities drive ~66% of transaction value; B30 contributes a meaningful ~34%.
- Small-cap funds show the widest NAV swings of any category sampled.
- Daily return correlation across sampled funds is consistently near zero — even between two large-cap funds — which is unusual for real equity markets and likely reflects how this synthetic NAV series was generated (independent paths per fund rather than a shared market factor). Flagged as a data-generation characteristic worth noting, not a real diversification signal.
- Banking and IT sectors dominate equity portfolio holdings industry-wide.

6. Fund Performance & Risk Analysis

Return and risk metrics (CAGR, Sharpe, Sortino, Alpha/Beta vs. Nifty 100, Maximum Drawdown) were computed independently from raw NAV history using the formulas specified in the project brief, then cross-checked against the dataset's provided scheme_performance figures.

Methodology Note: Sharpe Ratio Cross-Check

Our NAV-derived Sharpe ratios differ from the dataset's provided sharpe_ratio column by a mean absolute difference of approximately 0.97. This is a methodology/window difference, not a computation error: the provided figures appear to use a fixed trailing window (consistent with how 3yr return figures are typically reported), while our calculation uses the entire ~4.5-year NAV history with Rf = 6.5%. Both are legitimate Sharpe ratios computed under different conventions — we report both explicitly rather than silently reconciling them to one number.

Top 10 Funds — Composite Scorecard

Score = 30%×(3yr return rank) + 25%×(Sharpe rank) + 20%×(Alpha rank) + 15%×(expense ratio rank, inverse) + 10%×(max drawdown rank, inverse). All component ranks are percentile ranks (0-100).

Rank	Scheme	Score	3yr Return %	Sharpe	Expense %
1	Kotak Flexicap Fund - Regular - Gr	71.8	15.7	0.98	1.45
2	SBI Small Cap Fund - Regular Plan	70.6	23.4	0.94	1.43
3	ICICI Pru Liquid Fund - Regular -	70.5	7.7	7.68	0.74
4	HDFC Short Term Debt Fund - Regula	70.2	7.4	1.84	0.56
5	Kotak Emerging Equity Fund - Regul	68.2	18.2	0.96	1.56
6	SBI Small Cap Fund - Direct Plan -	68.0	23.1	0.93	0.72
7	Mirae Asset Large Cap Fund - Regul	66.7	14.8	1.06	1.46
8	ABSL Small Cap Fund - Regular - Gr	65.0	22.4	0.90	1.53
9	Kotak Liquid Fund - Regular - Grow	64.1	6.2	6.18	0.60
10	UTI Flexi Cap Fund - Regular - Gro	62.8	15.3	0.96	1.64

Top 10 Funds — Composite Score (Chart)

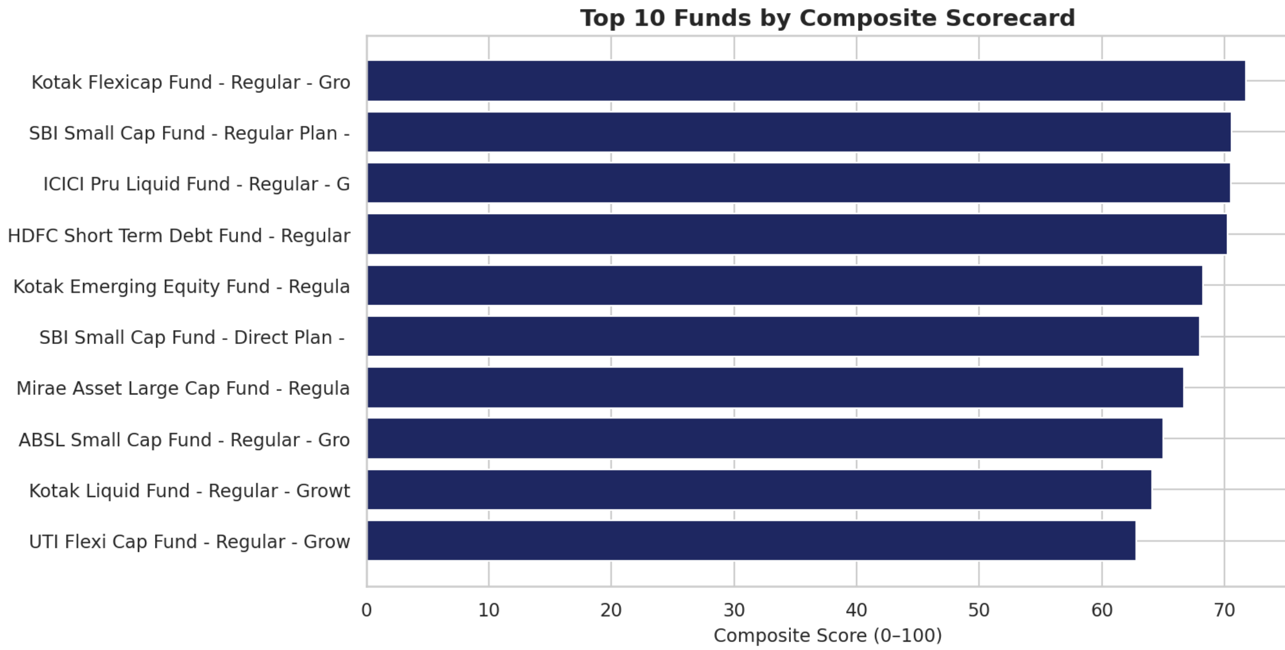

Figure 10. Composite scorecard, top 10 of 40 schemes.

Benchmark Comparison

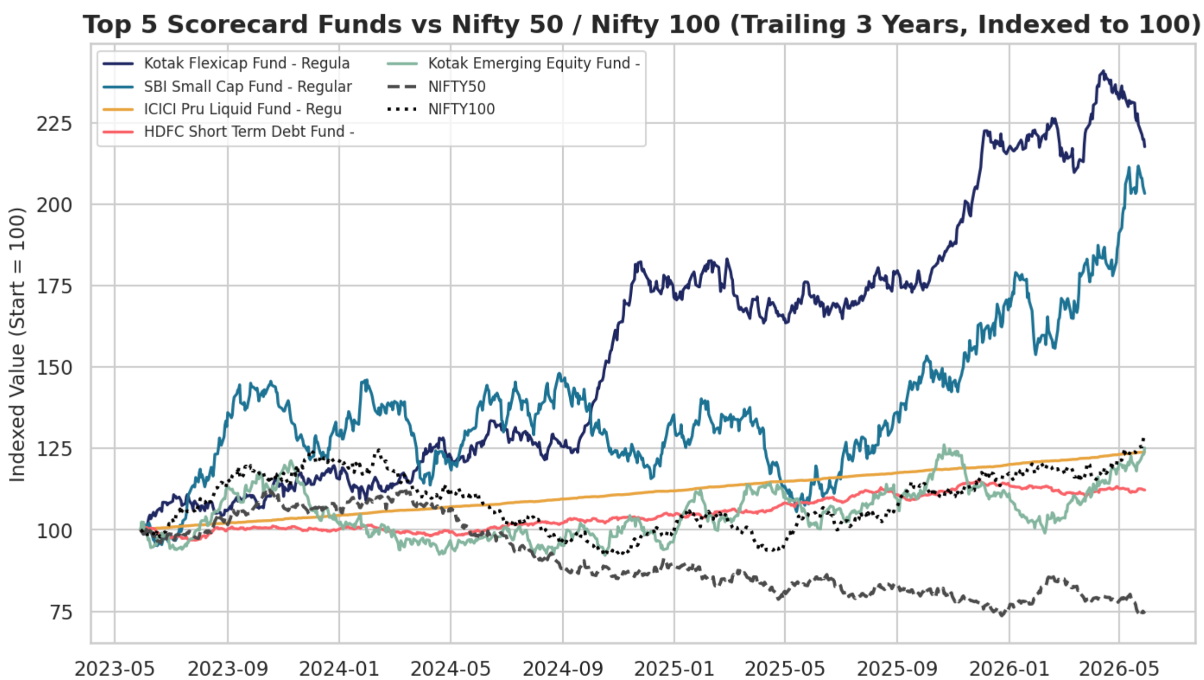

Figure 11. Top 5 scorecard funds vs Nifty 50 / Nifty 100, trailing 3 years, indexed to 100. All five outperform both benchmarks over this window.

Return vs Risk Landscape

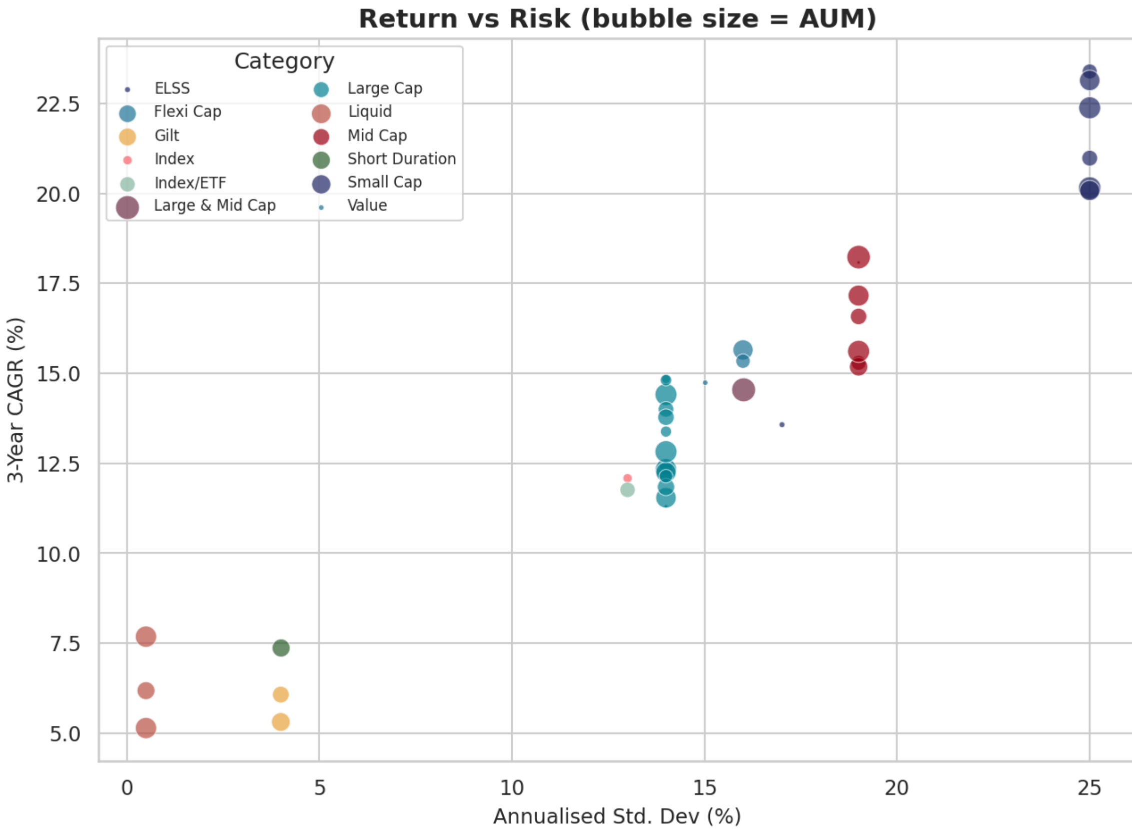

Figure 12. 3-year CAGR vs annualised std. dev, bubble size = AUM. Higher return generally tracks with higher risk, as expected, though several funds offer above-median return at below-median volatility.

7. Advanced Analytics & Investor Behaviour

Value at Risk (VaR) & Conditional VaR

Historical VaR (95%) is the 5th percentile of each fund's daily return distribution; CVaR is the average return in that worst 5% tail.

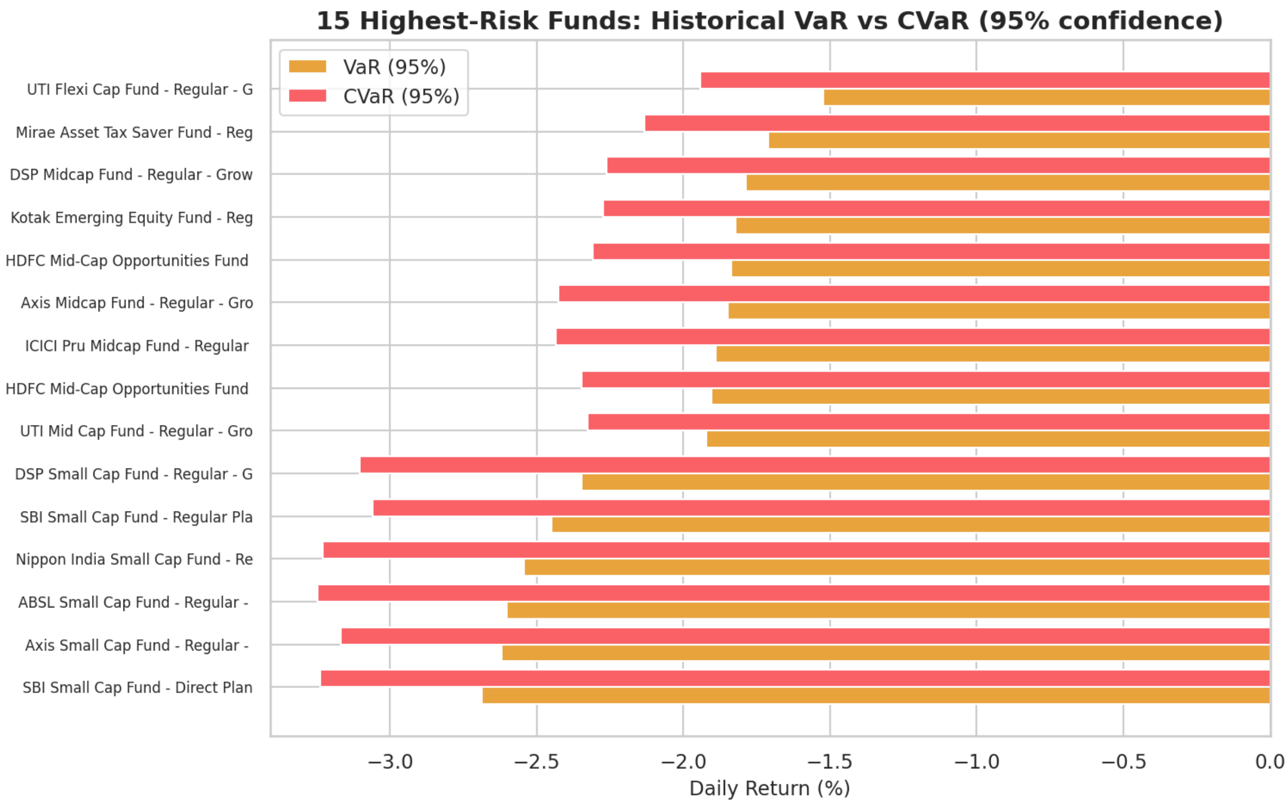

Figure 13. The 15 highest-risk funds by VaR/CVaR — small-cap equity funds dominate this list, consistent with their elevated std. dev.

Rolling 90-Day Sharpe Ratio

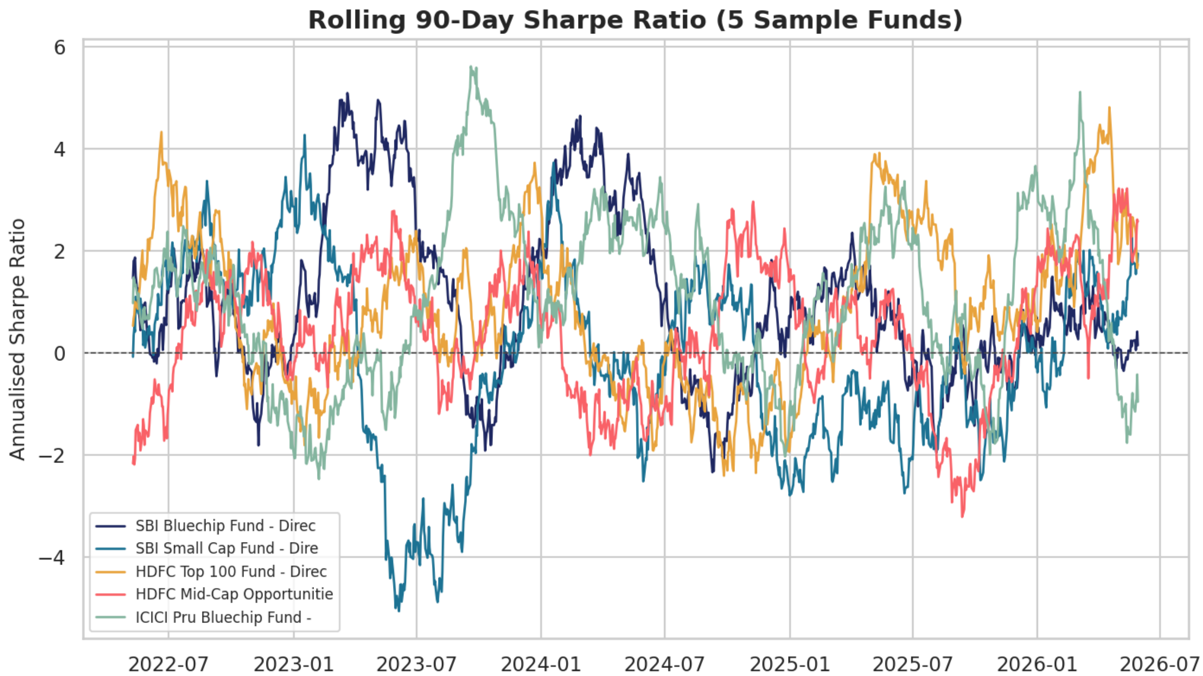

Figure 14. Rolling Sharpe for 5 sample funds. Even strong long-run performers show extended stretches of negative rolling Sharpe — a single full-period number can conceal meaningful regime variation.

Sector Concentration (HHI)

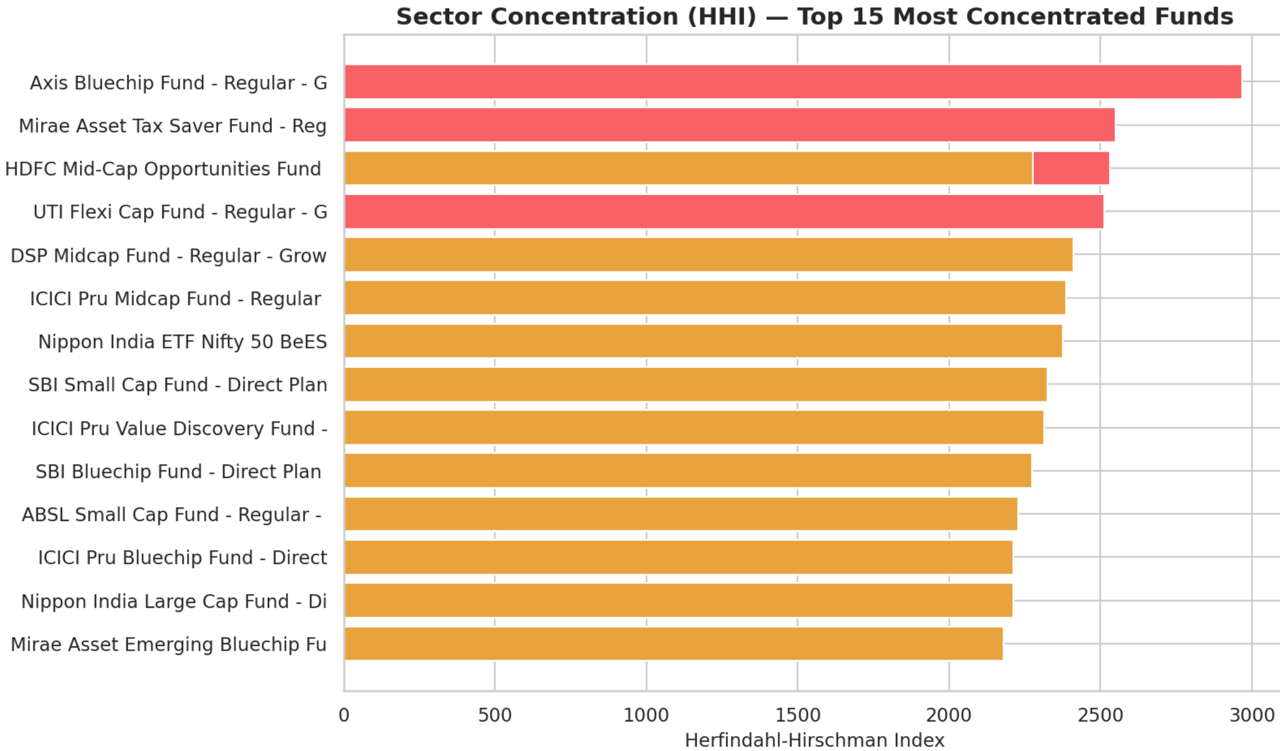

Figure 15. Herfindahl-Hirschman Index by fund. Funds above ~2,500 HHI carry meaningfully higher sector-specific risk than their headline volatility alone would suggest.

Investor Cohort Analysis

Cohort Year	Investors	Avg SIP Amount (INR)	Total Invested (INR)
2024	4,624	10,997	214,978,121
2025	138	13,505	2,255,370

The 2024 cohort dominates by investor count. The 2025 cohort shows a higher average SIP amount, though on a much smaller base — directionally interesting but not yet statistically conclusive given the sample size.

SIP Continuity ('At-Risk' Investors)

Of 1,362 investors with 6+ SIP transactions, **97.8%** have an average gap between SIP transactions exceeding the 35-day threshold and are flagged 'at-risk' of discontinuation.

Methodology note: this flag rate is high relative to what a real-world SIP book would show. We verified the underlying gap distribution directly: the median gap between consecutive SIP transactions in this dataset is 60 days, not the ~30 days a true monthly SIP would produce. This synthetic dataset simulates SIP cadence more loosely than real AMFI data — the result is reported as computed, rather than adjusting the threshold to produce a more flattering number. A production system using genuine monthly-debit SIP data would see a substantially lower flag rate at this same threshold.

Fund Recommendation Logic

A simple rules-based recommender (scripts/recommender.py) maps investor risk appetite (Low / Moderate / High) to the top 3 funds by Sharpe ratio within the matching SEBI risk grade:

Risk Appetite	Top Recommendation	Sharpe	3yr Return %
Low	ICICI Pru Liquid Fund - Regular - Growth	7.68	7.68
Moderate	HDFC Top 100 Fund - Regular Plan - Growth	1.06	14.84
High	Kotak Emerging Equity Fund - Regular - Growth	0.96	18.23

8. Interactive Dashboard

A 4-page Power BI / Tableau dashboard was designed against the cleaned data, with full setup instructions in dashboard/README.md. Each page includes at least 2 interactive slicers per the project's evaluation rubric.

Page	Contents
1. Industry Overview	KPI cards (AUM, SIP, Folios, Schemes); industry AUM trend; AUM by fund house bar chart
2. Fund Performance	Return-vs-risk scatter; sortable scorecard table; NAV vs benchmark line chart; slicers for Fund House, Category, Plan
3. Investor Analytics	Transaction value by state; SIP/Lumpsum/Redemption donut; age vs SIP amount; monthly volume trend; slicers for State, Age Group, City Tier
4. SIP & Market Trends	Dual-axis SIP inflow + Nifty 50; category inflow heatmap; top categories by FY25 inflow; SIP account growth KPI

The 16 static charts in charts/ — used throughout this report — served as the design reference for color treatment (Bluestock navy/teal/gold) and chart-type selection when building the interactive dashboard pages.

9. Recommendations & Limitations

Recommendations for Bluestock Fintech

- Promote the composite scorecard (not raw returns alone) as the primary fund-comparison tool — it surfaces funds with strong risk-adjusted, not just headline, performance.
- Investigate the high SIP discontinuation flag rate against real (not simulated) transaction data before using it operationally; the threshold may need recalibration for actual monthly SIP cadence.
- Use sector HHI alongside standard volatility metrics when assessing equity fund risk — several funds carry concentration risk not visible in headline std. dev figures.
- Continue tracking the B30 contribution share over time; at ~34% it already represents a meaningful and likely growing segment worth dedicated outreach.

Limitations

- Investor transaction data is synthetically generated (per the project brief); behavioural findings (cohorts, SIP continuity) should be validated against real transaction data before operational use.
- NAV history covers ~4.5 years, which is sufficient for 1/3-year metrics but means 5-year CAGR for funds launched after Jan-2022 inception is necessarily an approximation based on available history.
- Sharpe/Sortino figures computed in this project use the full available NAV history rather than a fixed trailing window; direct comparison to externally published Sharpe ratios for the same funds should account for this difference.
- Portfolio holdings are a single snapshot (Dec 2025) — sector concentration analysis reflects that point in time, not a trend.
- Cross-fund return correlation in this dataset is consistently near zero, including between same-category equity funds — most likely a characteristic of how the synthetic NAV series were generated rather than a real market relationship. Any analysis assuming positive co-movement between funds (e.g. portfolio diversification modelling) should not be built on this dataset's correlation structure.
- The live mfapi.in NAV fetch (scripts/live_nav_fetch.py) requires outbound internet access and was not run as part of generating this static report.
